

Program - 08

Create kubernetes Pods using YAML Descriptors

Structure of the program:

- nodejs-configmap.yaml
- nodejs-pod.yaml
- server.js

Steps:

1. Create a [server.js](#) file

```
const http=require('http');
const port=3000;

const server=http.createServer((req,res)=>{
  res.end("Hello from the node.js inside the kubernetes");
});

server.listen(port,()=>{
  console.log(`Server running on port ${port}`);
});
```

2. Build the config map file

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require('http');
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from the node.js inside the kubernetes");
    });

    server.listen(port, () => {
      console.log(`Server running on port ${port}`);
    });

```

3. Build the pod yaml file

```

apiVersion: v1
kind: Pod
metadata:
  name: prog8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
        items:
          - key: server.js
            path: server.js

```

4. Apply the yaml files

```
Kubectl apply -f nodejs-configmap.yaml
```

```
Kubectl apply -f nodejs-pod.yaml
```

5. Check the status

```
Kubectl get pods
```

```
Kubectl get deployments
```

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program8$ kubectl apply -f nodejs-pod.yaml
pod/prog8 configured
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program8$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
mycron-29424430-btd7k	0/1	Completed	0	7h32m
mycron-29424435-c2pkj	0/1	Completed	0	7h27m
mycron-29424440-k67cj	0/1	Completed	0	7h22m
mycron-29424865-t555g	0/1	ImagePullBackOff	0	13m
mycron-29424870-w87w9	0/1	ImagePullBackOff	0	12m
mycron-29424875-9cpjd	0/1	ImagePullBackOff	0	7m45s
mycron-29424880-s2295	0/1	ImagePullBackOff	0	2m45s
nginx-deployment-6f9664446b-klpxh	0/1	ImagePullBackOff	0	7h47m
nginx-deployment-6f9664446b-pv7kl	0/1	ImagePullBackOff	0	7h47m
prog8	1/1	Running	1 (13m ago)	7h29m

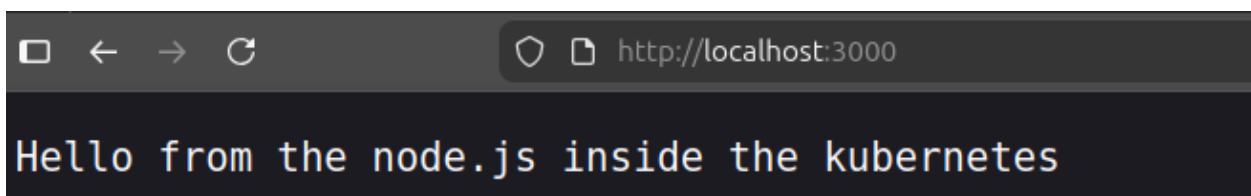
6. Access the web page

```
Kubectl port-forward nod/prog8 3000:3000
```

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program8$ kubectl port-forward pod/prog8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

7. Check the output in:

<http://localhost:3000>



8. Delete all the pods

Kubectl delete pods - -all

9. Delete all the services

Kubectl delete svc - -all

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program8$ kubectl delete services --all
service "kubernetes" deleted from default namespace
service "nginx-deployment" deleted from default namespace
```

10. Delete all the deployments

Kubectl delete deployments - -all

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/program7$ kubectl delete deployment --all
deployment.apps "nginx-deployment" deleted from default namespace
```